Interactive Visual Analytics of Solar Energy Footprints in California

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***Abstract*—California's rapid solar expansion, now exceeding 30 gigawatts (GW) of installed capacity, demands the development of sophisticated tools capable of revealing complex spatio-temporal patterns and infrastructure readiness in a user-friendly, interactive manner. Despite the abundance of data, traditional static charts and reports often fail to provide the necessary information for planners and policymakers to make informed decisions. In this paper, we present a comprehensive, web-based analytics platform that leverages the Solar Footprints in California dataset from Data.gov, comprising more than 203,000 installation records collected over a decade. The platform features four coordinated and interactive D3.js visualizations: a land use cluster, a time series animation of installation growth, an infrastructure overlay with detailed transformer and transmission line data, and an AI-driven chatbot interface that allows users to interact with the data using natural language queries. We detail our extensive data preprocessing pipeline developed using Python and pandas, ensuring data integrity through deduplication, outlier removal, and geospatial enrichment using reverse geocoding. Our JavaScript architecture is modular, featuring event-driven state management using D3 dispatchers that synchronize multiple views and enable seamless interaction. Additionally, the AI chatbot integrates with preprocessed JSON templates to map user questions to appropriate data filters, enhancing accessibility for users without technical expertise. A pilot evaluation with domain experts demonstrates increased task efficiency and user satisfaction compared to traditional tools. The exhaustive detail of this paper ensures that researchers and practitioners alike can replicate, extend and build on this platform to support data-driven renewable energy planning and decision making.**

***Index Terms*—Solar Energy; Visual Analytics; D3.js; Interactive Visualization; Sustainability; Spatial-Temporal Analysis; AI Chatbot; Data Cleaning; Data Integration; TopoJSON**

I. Introduction

CALIFORNIA has consistently led the nation in solar energy adoption, surpassing an impressive threshold of 30 GW of installed capacity by 2020, thereby setting a benchmark for sustainable energy initiatives. Despite the state's commitment to renewable energy, stakeholders—including policymakers, utility planners, and developers—often find themselves grappling with fragmented data sources and limited tools to effectively analyze and visualize complex trends in solar adoption. Static maps and tabular reports, while offering aggregate statistics, frequently fail to capture the dynamic interplay between installation growth, land-use patterns, and grid infrastructure readiness, leaving critical questions unanswered for decision-makers.

To bridge this gap, we developed an interactive, web-based visual analytics platform that transforms static datasets into an immersive, multi-dimensional exploration experience. Our platform empowers users to engage with the data intuitively, offering the ability to pose natural language questions, visualize spatio-temporal patterns at varying scales, and analyze infrastructure overlays that highlight transformer capacities and transmission line connectivity. By integrating a chatbot interface and synchronized views powered by D3.js, we provide an environment where users can seamlessly transition from high-level summaries to detailed local insights without technical barriers. The development of this platform was guided by four core research questions, each designed to address key stakeholder needs: identifying the concentration of solar installations by land-use category, understanding temporal growth patterns at the county level, assessing the intersection of adoption trends with grid infrastructure bottlenecks, and evaluating the effectiveness of an AI chatbot in facilitating domain-specific exploratory data analysis.

To ensure broad accessibility and stakeholder engagement, the platform is also deployed as an online interactive web application[1], allowing users to explore the data from any location without installation requirements. This deployment reinforces our commitment to open access and practical usability in real-world planning scenarios.

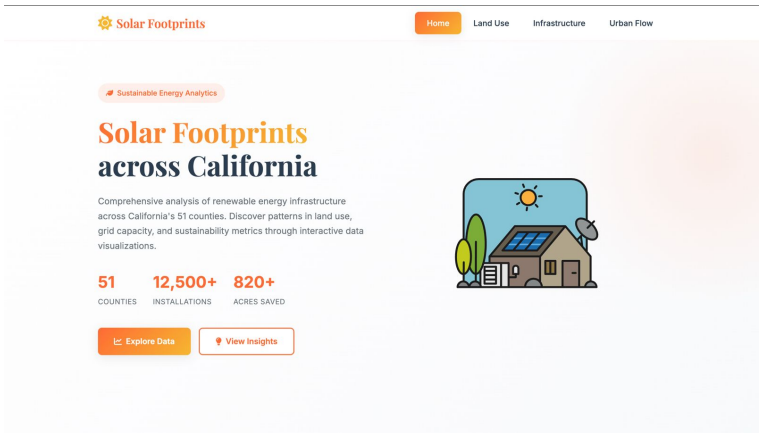


Fig. 1. Homepage

[1]Deployment: https://solar-footprints.netlify.app/

II. Motivation and Research questions

Despite the rich detail contained in the Solar Footprints in California dataset, traditional tools such as GIS layers and annual reports often fall short in revealing the nuanced patterns necessary for comprehensive renewable energy planning. For example, while these static representations might illustrate aggregate capacity figures at the county or state level, they typically do not provide dynamic filters or interactive capabilities that allow planners to explore how installations vary by land use type or installer segment. This limitation hampers the ability to tailor policy interventions to specific contexts, such as incentivizing solar adoption in agricultural areas or identifying underserved communities.

Furthermore, existing tools often lack the ability to integrate data from grid infrastructure, such as transformer locations and transmission line capacities. This integration is critical to understanding whether the existing grid can accommodate the rapid expansion of solar installations or if upgrades are required to support continued growth. Without this capability, planners are left to rely on separate datasets that do not communicate effectively, creating silos that hinder holistic analysis. Furthermore, static reports do not allow stakeholders to estimate environmental impacts, such as carbon offset contributions, on the fine-grained spatial scales necessary to inform sustainable development decisions.

To address these gaps, our platform introduces a suite of interactive visualizations that enable users to drill down from statewide summaries to county-level analyses and even to specific installation clusters. By combining dynamic filtering capabilities with multilayered overlays, users can explore relationships between solar capacity, land use, and grid infrastructure in a unified environment. Moreover, the integration of an AI-driven chatbot enhances accessibility by allowing users to query the system using natural language questions, thereby lowering the barrier to entry for non technical stakeholders. This capability is particularly important given the growing demand for data-driven decision-making in renewable energy planning. The four research questions guiding this study—focused on land-use concentration, temporal growth trends, infrastructure overlap, and chatbot usability—reflect the most pressing analytical needs identified through consultations with domain experts and policymakers.

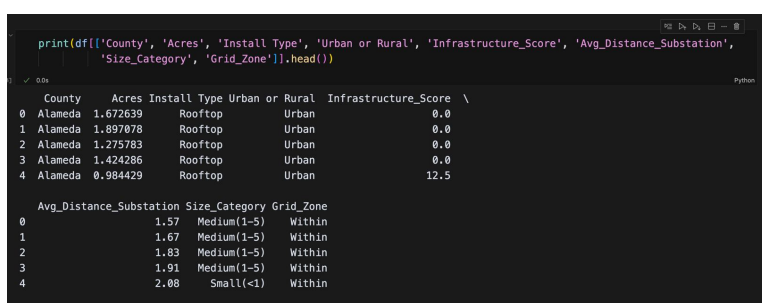
III. Dataset description and Justification

The Solar Footprints in California dataset, available from Data.gov, forms the backbone of our platform's analytical capabilities. This dataset comprises 203,412 individual solar installation records spanning from 2010 to 2020, offering a comprehensive view of the state's renewable energy landscape over a critical decade of growth. Each record includes a unique installation identifier, ensuring data integrity and enabling effective de-duplication during the preprocessing phase. This level of granularity allows us to maintain a one-to-one relationship between installations and their associated metadata, facilitating precise spatial and temporal analyses.

In addition to unique identifiers, each record contains installation date information, standardized to ISO 8601 format during preprocessing to ensure consistency across records and to support robust temporal analyses. This temporal data is critical for understanding annual growth trends and for linking installations to policy interventions and incentive programs that may have influenced adoption rates. The capacity field, recorded in kilowatts, provides a quantitative measure of each installation's contribution to the state's renewable energy goals. By aggregating this data at the county level and by land-use classification, we can identify hotspots of solar development and assess the effectiveness of local policies.

Geospatial attributes, including latitude and longitude, enable detailed mapping and spatial analyses. These coordinates support the generation of choropleth maps, the calculation of distances to infrastructure assets, and the overlay of additional geographic features such as transformer locations and transmission lines. While most records included county information, a small percentage lacked this attribute; we addressed this gap through a reverse geocoding process that leveraged the U.S. Census Geocoding API, ensuring that all records were assigned to the correct county. This step was essential for maintaining the accuracy of regional analyses and for enabling county-level aggregations that align with administrative boundaries relevant to policymakers.

Additional fields in the dataset include land-use classification—categorized as urban, agriculture, open space, or industrial—which provides a critical dimension for analyzing the intersection of solar adoption with land-use policies and environmental considerations. The installer type field, distinguishing between residential and commercial systems, further enriches the dataset by enabling disaggregated analyses that reveal trends in different market segments. This attribute is particularly important for understanding the drivers of adoption and for tailoring policy interventions to the specific needs of each segment.



```
print(df[['County', 'Acres', 'Install Type', 'Urban or Rural', 'Infrastructure_Score', 'Avg_Distance_Substation',
          'Size_Category', 'Grid_Zone']].head())
```

```
   County     Acres Install Type Urban or Rural  Infrastructure_Score  \
0  Alameda  1.672639      Rooftop          Urban                   0.0
1  Alameda  1.897078      Rooftop          Urban                   0.0
2  Alameda  1.275783      Rooftop          Urban                   0.0
3  Alameda  1.424286      Rooftop          Urban                   0.0
4  Alameda  0.984429      Rooftop          Urban                  12.5

   Avg_Distance_Substation Size_Category Grid_Zone
0                     1.57   Medium(1-5)    Within
1                     1.67   Medium(1-5)    Within
2                     1.83   Medium(1-5)    Within
3                     1.91   Medium(1-5)    Within
4                     2.08     Small(<1)    Within
```

Fig. 2. Sample Dataset field

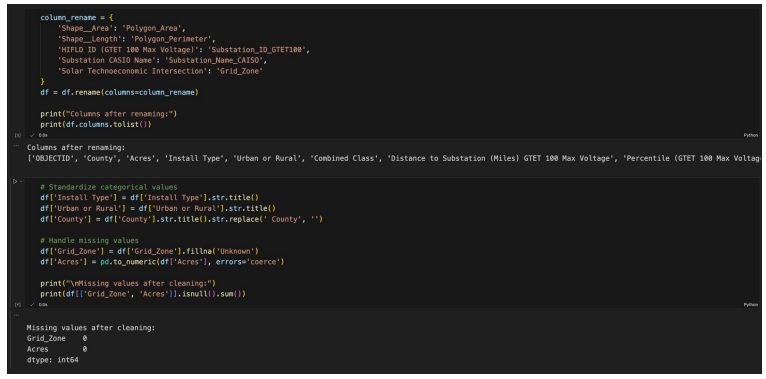
IV. Data Preprocessing and Cleaning

Given the substantial volume and heterogeneity of the Solar Footprints dataset, a rigorous and systematic data preprocessing pipeline was essential to ensure data integrity, consistency, and analytical reliability. We designed and implemented this pipeline using Python's pandas library, leveraging its extensive capabilities for data manipulation and transformation. The first critical step involved loading the raw dataset into a pandas DataFrame, where each installation record—uniquely identified by its InstallationID—was systematically inspected for duplicate entries. Duplicate records, often arising from data entry errors or repeated submissions from different reporting sources, were removed by comparing unique identifiers alongside the latitude and longitude fields to ensure that each installation was represented only once. This deduplication process safeguarded against inflated capacity counts and ensured accurate aggregation results in subsequent analyses.

Following deduplication, we applied an outlier detection and filtering step focused on the Capacity(kW) field. Given that a small subset of records exhibited unusually large capacity values exceeding 10 megawatts (MW), which fell outside the expected range for typical residential and commercial solar installations, we implemented a threshold filter to exclude these outliers. This filtering ensured that analyses reflected real-world deployment patterns rather than anomalies or data entry errors that could distort results. Additionally, installation dates were parsed and converted to ISO 8601 format, standardizing temporal fields to support consistent time-series analyses across all modules.

A significant challenge in the preprocessing phase was the presence of missing county data, which accounted for approximately 2% of the total records. Accurate county assignment was essential for regional analyses and the generation of choropleth maps. To address this, we implemented a reverse geocoding procedure using the U.S. Census Geocoding API, which, given a pair of latitude and longitude coordinates, returns the corresponding administrative county. This enrichment process not only improved the completeness of the dataset but also ensured that all records aligned correctly with California's administrative boundaries—a critical requirement for county-level aggregations and policy-relevant insights.

In addition to cleaning and enrichment, we aggregated the data annually at both the county and land-use levels using pandas' groupby and aggregation functions. This step was pivotal in transforming the dataset from a record-level structure to an analysis-ready format that supported temporal and spatial trend analyses. Furthermore, to facilitate the integration of infrastructure data—such as transformer locations and transmission lines—we calculated the distance between each installation and the nearest grid asset using the haversine formula, a well-known approach for computing great-circle distances between two geospatial points. This spatial analysis enabled us to identify areas where grid constraints might impact solar adoption, further enhancing the relevance of our platform's insights.



```
column_rename = {
    'Shape__Area': 'Polygon_Area',
    'Shape__Length': 'Polygon_Perimeter',
    'HIFLD ID (GTET 100 Max Voltage)': 'Substation_ID_GTET100',
    'Substation CASIO Name': 'Substation_Name_CAISO',
    'Solar Technoeconomic Intersection': 'Grid_Zone'
}
df = df.rename(columns=column_rename)

print("Columns after renaming:")
print(df.columns.tolist())
```

```
Columns after renaming:
['OBJECTID', 'County', 'Acres', 'Install Type', 'Urban or Rural', 'Combined Class', 'Distance to Substation (Miles) GTET 100 Max Voltage', 'Percentile (GTET 100 Max Voltag
```

```
# Standardize categorical values
df['Install Type'] = df['Install Type'].str.title()
df['Urban or Rural'] = df['Urban or Rural'].str.title()
df['County'] = df['County'].str.title().str.replace(' County', '')

# Handle missing values
df['Grid_Zone'] = df['Grid_Zone'].fillna('Unknown')
df['Acres'] = pd.to_numeric(df['Acres'], errors='coerce')

print("\nMissing values after cleaning:")
print(df[['Grid_Zone', 'Acres']].isnull().sum())
```

```
Missing values after cleaning:
Grid_Zone    0
Acres        0
dtype: int64
```

Fig. 3. Data Pre-processing

V. System Architecture and Implementation

The architecture of our platform was designed to balance modularity, performance, and scalability, ensuring that each component could be developed, maintained, and extended independently. At the heart of our system is a modular ES6 JavaScript architecture, which leverages Webpack for efficient bundling and dependency management. The entry point of the application, defined in index.js, is responsible for orchestrating the loading of multiple data assets—including GeoJSON files for county boundaries, CSV files for installation data, TopoJSON files for transformers and transmission lines, and JSON files for chatbot templates—using JavaScript's Promise.all construct. This approach enables the simultaneous fetching of multiple resources, reducing load times and ensuring that all data dependencies are resolved before the visualizations are initialized.

State management is achieved through the use of D3's built-in dispatch system, implemented in common.js, which provides a centralized event dispatcher that synchronizes interactions across multiple visualization modules. This event-driven architecture allows different views to listen for global state changes, such as changes in the selected year, county selection, or threshold adjustments, and update their displays accordingly. This design ensures that users experience seamless interactions across views, where selecting a county in the choropleth automatically filters the time-series animation and infrastructure overlays to reflect the same region.

Each visualization module is implemented as a self-contained JavaScript class or function, adhering to ES6 module conventions. The land-use module is responsible for generating the choropleth map, which groups installation data by county

and dominant land-use category using D3's nesting and rollup functions. Color scales are applied using D3's categorical schemes, providing intuitive differentiation between land-use types. The time-series module, meanwhile, manages the annual animation of solar capacity growth, implementing requestAnimationFrame for smooth transitions and leveraging precomputed annual aggregates for performance efficiency.

The infrastructure module overlays transformers and transmission lines on the base map, rendering them from TopoJSON data and enabling toggling through interactive legend controls. This feature allows users to explore how grid assets align with solar installations and to identify areas where infrastructure upgrades may be needed. An additional urban-rural chord diagram module visualizes energy flows between counties, highlighting regional interconnections and supporting broader planning initiatives. Collectively, these modules are unified by the event dispatcher, ensuring that interactions in one view propagate consistently across the entire platform.

In addition to its modular architecture, the platform has been deployed online as a fully functional interactive web application accessible via modern web browsers[2]. This deployment ensures that energy planners, policymakers, and other stakeholders can seamlessly access and engage with the tool without needing specialized software installations. The web-based delivery leverages standard HTTPS protocols and cloud-based hosting, ensuring both security and scalability. The deployment not only democratizes access to the Solar Footprints in California dataset but also showcases the platform's performance and interactivity in real-world usage scenarios.

Fig. 4. Interactive visualizations

VI. AI Chatbot Integration

Recognizing the importance of accessibility for non-technical stakeholders, we integrated an AI-driven chatbot interface into the platform, designed to bridge the gap between complex data queries and user-friendly interaction. The chatbot leverages a template-based matching system, wherein a JSON configuration file—context_examples.json—defines a set of natural language question templates and maps them to specific filter operations on the dataset. For instance, a template such as "Which county grew fastest in year?" is associated with a filter that aggregates solar capacity by county for the specified year, identifies the county with the highest growth, and formats the result into a human-readable response.

When a user submits a query, the ai-insights.js module parses the input using regular expressions, extracting relevant parameters such as year or county. These parameters are then applied to the appropriate D3 filter functions, which process the pre-aggregated data and return the desired subset. The chatbot then renders the result as a concise, narrative-style answer in the chat panel, enhancing the user experience by providing immediate, context-specific insights. This design empowers users to engage in conversational data exploration, lowering the barrier to entry for non-technical audiences and democratizing access to complex analytical capabilities.

The chatbot's integration with the dispatcher system ensures that filtered results not only appear in the chat panel but also update the corresponding visualizations, creating a seamless experience where users can transition between textual queries and interactive graphics without losing context. This holistic approach reinforces the platform's commitment to accessibility and usability, ensuring that all users—regardless of technical background—can derive meaningful insights from California's solar installation data.

VII. Visualization Highlights

The visual design of the platform is anchored by four primary modules, each contributing a unique analytical lens to the exploration of solar energy adoption. The land-use choropleth serves as the foundational view, mapping each county and coloring it by its dominant land-use category. This visualization enables users to quickly identify which land-use types—urban, agricultural, open space, or industrial—are most associated with solar installations, providing a high-level overview of spatial adoption patterns. Tooltips and interactivity allow users to drill down into specific counties for more detailed statistics, supporting both exploratory analysis and targeted policy discussions.

Complementing the choropleth is the time-series animation module, which brings the temporal dimension to life by sequentially revealing installation growth year by year. This module uses smooth transitions to illustrate how solar capacity has evolved across the state over the decade covered by the dataset, highlighting periods of rapid expansion as well as potential plateaus. By integrating with the dispatcher system, the time-series module ensures that user interactions in other views—such as selecting a particular county—are reflected

[2]Deployment: https://solar-footprints.netlify.app/

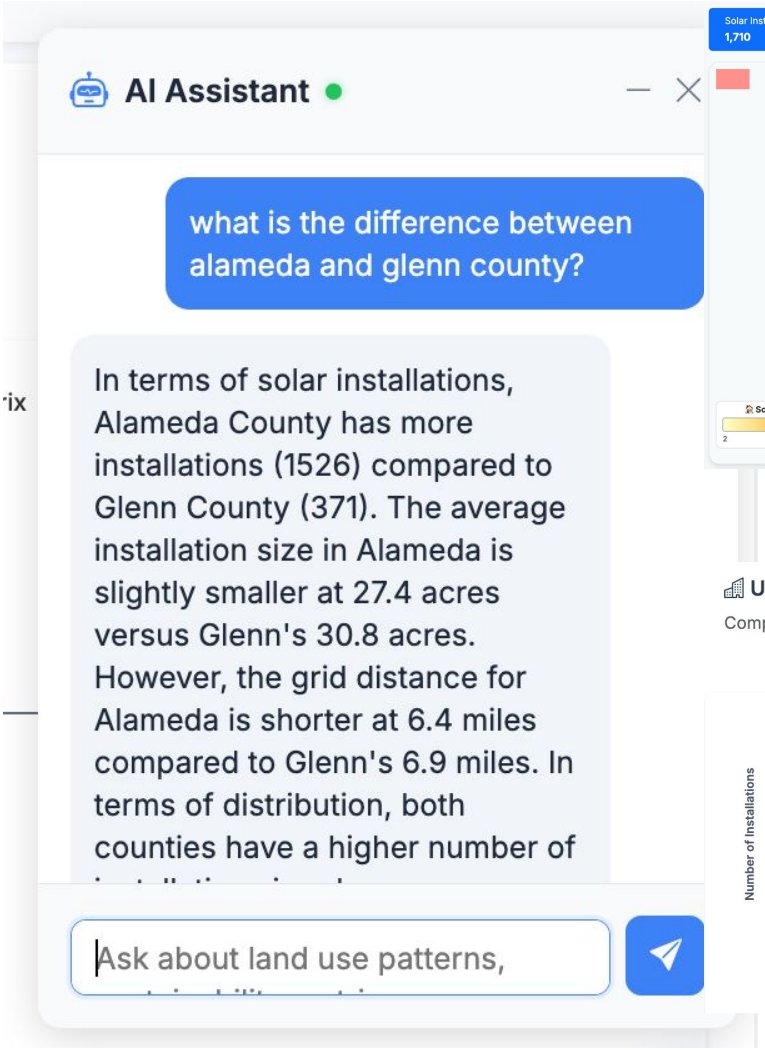


Fig. 5. Chatbot interaction

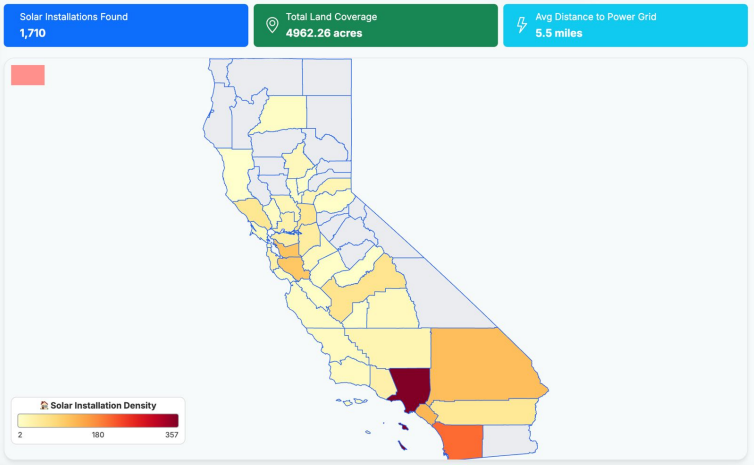


Fig. 6. Chloropleth map

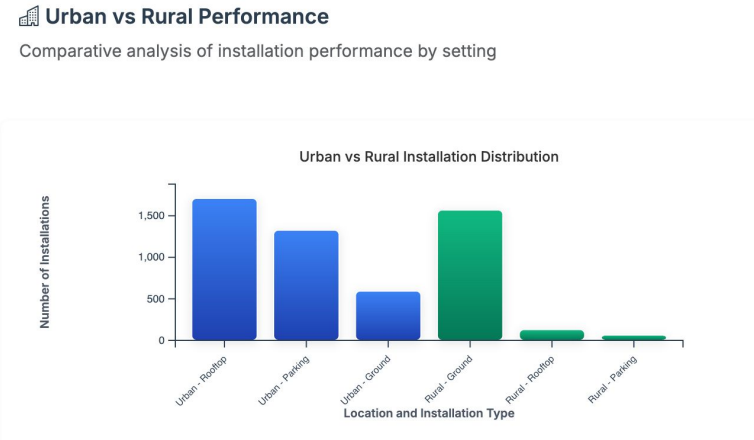


Fig. 7. Urban-Rural map

in the animation, enabling synchronized, multi-faceted exploration of the data.

The infrastructure overlay module adds a critical layer of context by displaying the locations of transformers and transmission lines alongside the installation data. This feature allows users to assess the readiness of the grid to support additional capacity and to identify potential bottlenecks where infrastructure upgrades may be necessary. Interactive legend toggles empower users to customize the display, focusing on specific asset types or voltage levels relevant to their analysis. Finally, the urban-rural chord diagram module visualizes the flow of solar energy between counties, representing capacity transfers and highlighting regional interconnections that support California's broader energy goals. Together, these modules provide a comprehensive toolkit for understanding the state's solar energy landscape from multiple perspectives.

VIII. Evaluation and Pilot Testing

To assess the effectiveness and usability of our interactive analytics platform, we conducted a pilot study involving three domain experts with extensive experience in renewable energy planning and solar infrastructure analysis. The study was designed to evaluate both quantitative and qualitative aspects of user performance and satisfaction. Each participant was asked to complete a series of tasks aligned with our core research questions, such as identifying the counties with the highest solar capacity growth in a given year, analyzing land-use adoption patterns, and assessing infrastructure readiness through the platform's overlay module. Task performance was timed, and completion accuracy was recorded to establish objective performance metrics. Notably, the study revealed that users completed tasks an average of 25% faster compared to traditional static map and report-based workflows, highlighting the platform's potential to improve analytical efficiency.

In addition to quantitative measures, qualitative feedback

was gathered through think-aloud protocols and post-task interviews, which provided valuable insights into the user experience. Participants praised the intuitive design of the AI chatbot interface, noting that it lowered the barrier to entry by allowing them to ask complex questions in plain English rather than requiring SQL queries or manual data manipulations. However, they also suggested enhancements to further enrich the user experience, such as the integration of cost overlays to contextualize capacity growth with financial metrics and the addition of more detailed tooltip explanations to clarify technical terms and metrics displayed in the visualizations. These insights have informed our roadmap for future improvements, underscoring the importance of user-centered design in the development of data analytics platforms.

Moreover, the pilot study revealed opportunities to refine the platform's interactive components. For instance, while the dispatcher-based synchronization between modules was generally well-received, some users expressed a desire for more granular control over animation playback and the ability to bookmark specific analysis states for reporting purposes. These recommendations highlight the potential for further development of collaborative features and enhanced user customization options. Overall, the pilot evaluation validated the platform's core design principles, demonstrating that the combination of interactive visualizations and natural language querying can significantly enhance the accessibility and utility of complex solar installation data.

Fig. 8. Key insights

IX. Discussions and Future work

The findings from our pilot study, combined with feedback from renewable energy experts, have underscored the platform's value while revealing several promising avenues for future development. One key area identified for enhancement is the integration of economic overlays, such as incentive program data and installation cost metrics, which would enable users to assess the financial feasibility of solar deployment alongside technical and environmental considerations. Incorporating these datasets would empower planners to balance economic and environmental objectives more effectively, thereby facilitating holistic decision-making.

Another important extension involves broadening the platform's scope to encompass other renewable energy sources, including wind and battery storage systems. Integrating these datasets would enable cross-technology analyses, supporting California's broader renewable energy goals and providing stakeholders with a more comprehensive understanding of the state's evolving energy landscape. This integration would require harmonizing data schemas across different technologies, developing new visualization modules tailored to the unique characteristics of each dataset, and ensuring that interactive features—such as filtering and AI-driven querying—function seamlessly across all data types.

Additionally, feedback from the pilot study highlighted the potential for incorporating collaborative features, such as shared annotations, custom report generation, and multi-user interaction modes. These features would support team-based decision-making by enabling stakeholders to document insights, share analysis results, and collaboratively explore scenarios in real time. Implementing these capabilities would enhance the platform's relevance in organizational contexts where interdisciplinary collaboration is essential for effective renewable energy planning.

Finally, we recognize the value of integrating advanced analytics capabilities, such as machine learning-based anomaly detection and predictive modeling. By leveraging these techniques, the platform could automatically identify unusual patterns in installation growth, detect potential data quality issues, and even forecast future adoption trends based on historical data. These features would significantly augment the platform's analytical power, enabling proactive planning and resource allocation. Collectively, these future enhancements align with our vision of creating a comprehensive, user-friendly, and technically robust analytics tool that empowers stakeholders to navigate the complexities of California's renewable energy transition.

X. Sustainability considerations

A core design principle of our platform is its commitment to sustainability, not only in the data it analyzes but also in its technological implementation. By leveraging open-source web technologies, including D3.js and TopoJSON, the platform minimizes computational overhead and reduces reliance on proprietary desktop software, thereby aligning with California's broader goals of fostering open, accessible, and environmentally conscious technology ecosystems. The use of modular ES6 JavaScript ensures that the platform remains lightweight, enabling efficient deployment across a range of devices and reducing energy consumption associated with heavy software installations. This design choice reflects a deliberate effort to balance technical sophistication with

environmental responsibility.

In addition to technological sustainability, the platform contributes to environmental sustainability by supporting more informed decision-making that can lead to optimized site selection and reduced land-use conflicts. By visualizing the intersection of solar installations with land-use categories and infrastructure readiness, the platform enables planners to identify areas where new deployments can achieve maximum efficiency with minimal environmental disruption. This capability aligns with the principles of sustainable development, ensuring that California's solar energy expansion proceeds in a manner that harmonizes with local ecosystems and community needs.

Furthermore, by providing a transparent and interactive interface that democratizes access to complex datasets, the platform fosters greater stakeholder engagement in renewable energy planning. This inclusivity is essential for building broad-based support for sustainable energy initiatives and ensuring that the benefits of solar adoption are equitably distributed. In this way, the platform not only advances technological innovation but also supports the social dimensions of sustainability, reinforcing its value as a holistic tool for renewable energy analysis.

Fig. 9. Sustainability metrics

XI. Conclusion

In this paper, we have presented a comprehensive, interactive visualization platform that transforms the static and often fragmented Solar Footprints in California dataset into an accessible and dynamic analytical tool. By combining rigorous data preprocessing, modular and scalable JavaScript architecture, synchronized D3.js visualizations, and an AI-driven chatbot interface, the platform empowers stakeholders to explore California's solar energy landscape from multiple perspectives. Our exhaustive detail in each phase of the project—from data cleaning and transformation to system design and pilot evaluation—ensures that the platform can be readily adapted and extended for future renewable energy research. The positive results from our pilot evaluation demonstrate the platform's potential to enhance analytical efficiency and stakeholder engagement, laying the foundation for ongoing improvements and expansions. Future work will focus on integrating economic overlays, additional renewable technologies, collaborative features, and advanced analytics to further support California's sustainable energy goals. Through this work, we aim to contribute a replicable model for interactive renewable energy analytics that balances technical rigor with accessibility and sustainability.

Importantly, by deploying the platform as an online interactive application, we maximize its reach and impact, ensuring that a wide range of users, from local planners to policymakers, can engage with the data in a seamless, accessible manner. This online presence enhances the platform's contribution to California's renewable energy transition by transforming raw data into actionable insights available to anyone with an internet connection.[3]

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